Kyle R. Robin

Stowe, VT | krobin@bates.edu | linkedin.com/in/kyle-robin-931415425 | krobin2005.github.io/myportfolio | U.S. Citizen

Education

Bates College — Lewiston, ME
B.S., Major: Digital & Computational Studies (declared); Intended Second Major: Psychology. GPA: 3.71 — *Expected May 2029*
Relevant Coursework: Intro CS: Software Development (Python), Data Structures & Algorithms*, Evolutionary Computing*, Principles of Psychology* *(*in progress)*

Northeastern University, The Roux Institute (Graduate School) — Portland, ME
Graduate course, taken as an undergraduate through Bates: CY 5010 Cybersecurity Principles and Practices — *Spring 2027*

Burke Mountain Academy — East Burke, VT
High School Diploma. GPA: 4.00 — *May 2024*

Experience

Research Assistant (Neuroscience), Lead of AI Research — September 2026 – Present
Castro Lab, Bates College — *Lewiston, ME*

- Lead design and development of AI for Smell under Professor Jason Castro: a six-agent research lab in Python that trains neural networks to predict odor from molecular structure (5,858 molecules).
- Built a Skeptic agent that tests every claimed gain with scaffold splits, shuffled-label controls, and corrected confidence intervals.
- Added memory and evolutionary search; benchmarked against Lee et al. (*Science* 2023). Working toward a co-authored paper.

Technology Sector Analyst — August 2025 – Present
Bates Investment Club — *Lewiston, ME*

- Research technology companies (previously the industrial sector) and build investment theses using DCF, comparable-company, and fundamental analysis; present stock pitches to the club.

Greenskeeper — June 2024 – August 2026 (Summers)
West Hill Golf Course — *Stowe, VT*

- Led turf, irrigation, and course-design projects on a four-person grounds crew at a private golf course.
- Operated and maintained tractors, mowers, and sand-management equipment; applied fertilizer and chemicals under safety and environmental regulations.

Co-Lead Researcher and Project Analyst — January 2024 – May 2024
BMA Energy Proposal — *East Burke, VT*

- Co-led a $1.6M solar feasibility study covering 248,000+ kWh of annual campus electricity use, evaluating system sizing, project economics, and federal and Vermont incentives (IRA Elective Pay, ITC, PTC).

Projects

Strayline: Founder, AI Network-Attack Detection Venture — October 2026 – Present

- Designed a hybrid intrusion detector: a classifier for known attack families plus an autoencoder trained only on benign traffic, with an evaluation that maps which unseen attacks are caught at a 1% false-alarm rate.
- Submitted as a research proposal to the AAAI-27 Undergraduate Consortium; applied to pitch at The Eleven Society's Northeast Entrepreneurship Summit (Boston, Nov. 2026).

Scent to Art: Prize Winner, Bowdoin–Bates–Colby Hackathon — September 2026

- Built a five-agent Python pipeline on the Claude API that infers a molecule's chemical class from Pyrfume odor descriptors, proposes a structural change, and predicts the modified molecule's full odor profile.
- Grounded predictions with a batched web-search agent that falls back to a second dataset, and capped cost with a supervisor agent that halts each run at 30 API calls.
- Rendered results as radial "scent network" maps in Matplotlib and presented to ~100 attendees, including the president of Bowdoin College.

Athletics

NCAA Division I Alpine Skiing, Bates College — August 2025 – Present

- Balance 30+ hours per week of training and competition while maintaining a 3.71 GPA.

Internationally Competitive Ski Racer, BMA E-Team — May 2024 – April 2025

- Sponsored athlete in FIS, National Championship, and Continental Cup races across North America and Europe (Austria, Italy, Slovenia, Canada); earned FIS victories and top-10 National Championship results.

Skills

Programming: Python (pandas, NumPy, PyTorch, scikit-learn, RDKit, Matplotlib), R, HTML/CSS/JavaScript, Git and GitHub
AI: Claude API (multi-agent pipelines, tool use, web search), LLM prompting and evaluation, ChatGPT, Gemini, Codex
Finance: Financial accounting, DCF analysis, comparable-company analysis · **Software:** Excel, Word, PowerPoint
Certifications: Anthropic AI Fluency